

Symica: Circuit Design And Simulation Tools For All

Symica software suite makes IC design easy and accurate. By including various tools including schematic editor, simulator (using SPICE), waveform viewer and analyser, it provides a complete ecosystem to design and develop electronic systems.

Symica is compatible with Windows and Linux. It comes both as a free edition (FE) and paid version. Unlike full (paid) edition, its free edition does not support mixed-signal setup. However, all the other offerings make up for this drawback.

Features

Using schematic editor from Symica DE schematic development environment, designers can access the elements library to choose components and draw the circuit schematic. Other functionalities of the development environment include symbol editor and hierarchy editor. Designers can specify components using the component description format. Symica supports different netlist formats like Spice, LVS, HSpice and Spectre.

The created schematic is used by the SymSPICE circuit simulator to digitally run the circuit and show its behaviour. Developed by Symica's in-house R&D team, the SPICE simulation environment supports all industry-format external netlists as well as transistor models. It delivers very accurate simulation signals through a graphical interface. The free edition supports up to 300 elements for simulation and up to ten Verilog-A mode parameters.

The waveform viewer in Symica, called SymProbe, provides a graphical representation of the simulation waveform. It also helps in simulation result analysis. SymProbe tool provides view of results for DC, AC, SWEPT and

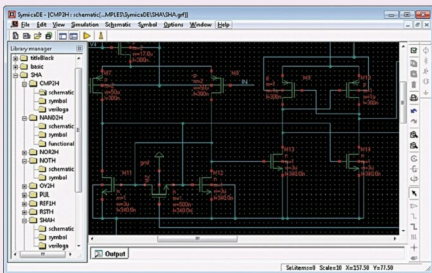


Fig. 1: A complete circuit schematic on Symica DE

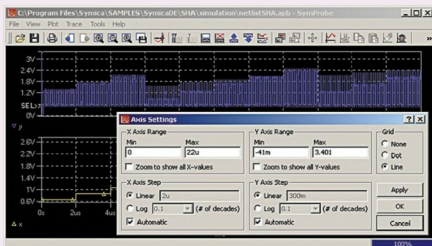


Fig. 2: Customising a SPICE simulation

TRAN; customisable view of multiple plots; multiple X/Y axes in one plot; and flexible axes adjustment. It can also measure cursor waveforms and brings enhanced printing facility with vector EMF waveform output. Simple drag-and-drop operations make it easy to use. SymProbe accepts input formats like APB, NUTMEG, CSV, CSDF and RAW-SPICE.

In addition, Symica includes a symbol generator for primitive devices (like diodes, transistors and MOS) relevant to the user's projects and a library translation tool that can import external libraries for use.

As regards feature upgrades, the software now allows current probing for complex transistor models and sub-circuits. There are two modes of